

AIR-GAPPED / ISOLATED STATEFUL SECURITY LABORATORY

Proof of Concept | Architecture, Build Procedure, Traffic Engineering, Telemetry, and Threat-Hunting Workflow



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Operational credentials from the supplied lab notes are intentionally redacted from this publication-ready document.

1. Executive Summary

This Proof of Concept documents an intentionally isolated, stateful security lab built to reproduce enterprise-style network segmentation, physical Layer 2 transport, controlled exploit paths, and high-fidelity packet telemetry. The design bridges infrastructure engineering and incident response: attack traffic is generated by a dedicated Kali boundary, forced through a physical Juniper SRX300 security policy boundary, carried across a Cisco Catalyst 2960-X 802.1Q trunk, delivered to a VLAN-tagged victim segment inside Proxmox, and simultaneously mirrored into a dedicated Security Onion capture interface.

The central engineering objective is not simply to make the virtual machines communicate. It is to make the traffic traverse real policy and switching boundaries so that captures, session logs, and detections reflect the physical path that would exist in a small enterprise environment. That creates a repeatable POC platform for packet analysis, Splunk/Elastic rule development, SIEM validation, DNS-exfiltration hunting, defensive control testing, etc.

Business / Engineering Value

- **High-fidelity telemetry:** physical routing and SPAN mirroring preserve realistic packet paths for capture and analysis.
- **Controlled blast radius:** vulnerable targets are isolated in VLAN 10 and only become reachable when the SRX policy permits the flow.
- **Stateful validation:** the lab demonstrates both permitted sessions and explicitly logged denied initiation attempts from the attacker boundary into management.
- **Reproducible engineering:** network, switching, virtualization, endpoint, and analytics steps are documented as a build sequence rather than an ad-hoc collection of settings.
- **Portfolio signal:** the POC demonstrates practical integration of Junos security policy, Cisco switching, Proxmox networking, Linux/Windows administration, and SOC analytics.

Publication Note

The source bundle includes operational usernames and passwords. Because this document is intended for LinkedIn publication, those credentials are not reproduced. All architecture, addresses, commands, interface bindings, and detection logic that can be safely shared are retained.

2. Lab Objectives and Design Principles

- **Create three distinct security domains:** ATTACKER, VICTIMS, and MGMT.
- **Force north-south and inter-zone traffic** through the Juniper SRX300 stateful boundary.
- Use Cisco 802.1Q trunking to transport VLAN 10 and VLAN 99 between the SRX and Proxmox environment.
- Use Proxmox vmbr0 as the VLAN-aware management/live-traffic bridge and vmbr1 as a dedicated SPAN/capture bridge.
- Mirror all VLAN 10 traffic from the Cisco switch to the Security Onion capture interface.
- Provide a Windows 11 dual-homed analyst station with an explicit static lab route but no Ethernet default gateway.
- Preserve a deterministic workflow from attack generation to packet capture, SIEM ingestion, and threat-hunting query execution.

Interpretation of "air-gapped": the lab is isolated from production and is intentionally segmented. The supplied topology also contains an external-boundary connection for the attacker network; therefore the design should be

treated as an isolated research range rather than a claim that every interface is physically disconnected from the Internet.

3. Reference Architecture

The topology is built around a physical Juniper SRX300, a managed Cisco Catalyst 2960-X, a Proxmox VE hypervisor, a Windows 11 analyst station, a Security Onion 3.1.0 sensor, Kali Linux, and an antiX Linux victim workload. The supplied topology diagram is the authoritative physical reference for the POC.

Updated Air-Gapped Security Lab Topology with Specific Physical Connections

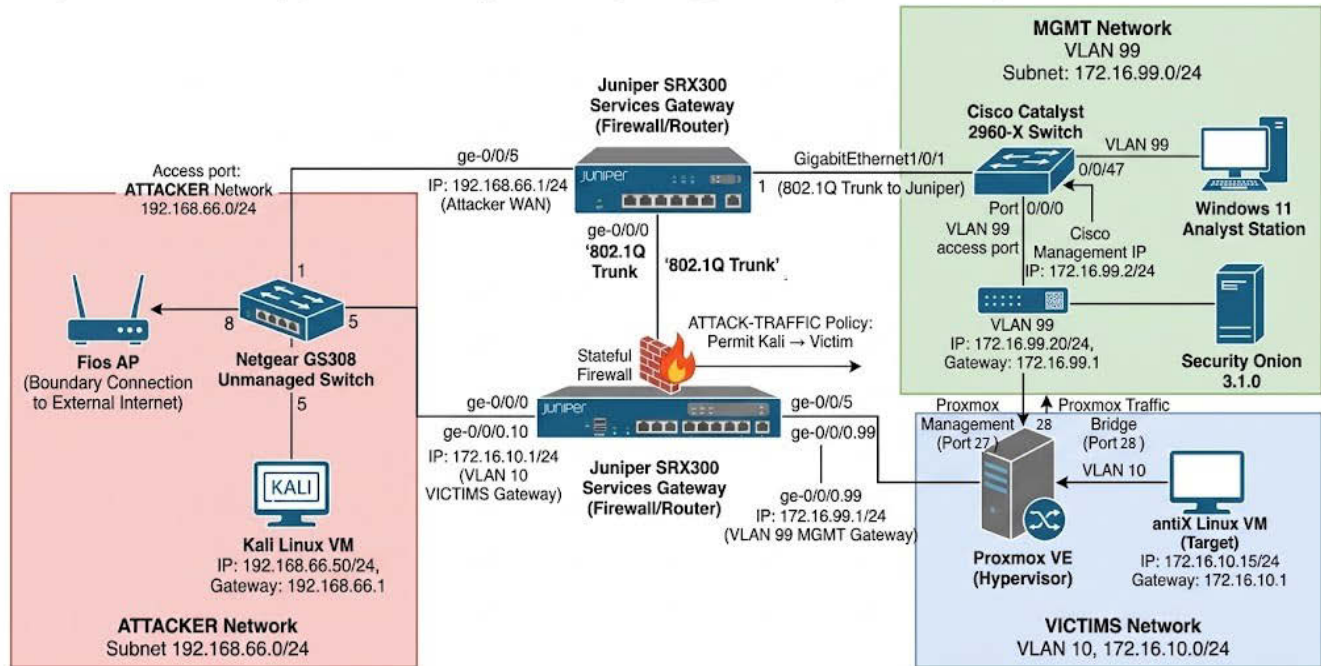


Figure 1. Updated lab topology and physical connections from the supplied design.

Security Domains and Address Plan

Domain	VLAN	Subnet	Gateway	Primary endpoints	Purpose
ATTACKER	Access / external boundary	192.168.66.0/24	192.168.66.1	Kali 192.168.66.50	Threat-emulation edge
MGMT	99	172.16.99.0/24	172.16.99.1	Windows 172.16.99.10; Proxmox 172.16.99.20; Cisco 172.16.99.2; Security Onion 172.16.99.30	Administration and SOC access
VICTIMS	10	172.16.10.0/24	172.16.10.1	antiX 172.16.10.15	Controlled blast radius

4. Physical Connection and Traffic Path

- Fios AP -> Netgear GS308: provides the boundary connection used to place the attacker network outside the internal management and victim domains.
- Netgear GS308 -> Juniper SRX300 ge-0/0/5.0: attacker segment gateway 192.168.66.1/24.
- Juniper SRX300 ge-0/0/0 -> Cisco Catalyst GigabitEthernet1/0/1: 802.1Q trunk carrying VLAN 10 and VLAN 99.
- Cisco GigabitEthernet1/0/27 -> Proxmox nic0/vmbr0: VLAN-aware management and live traffic path.
- Cisco GigabitEthernet1/0/28 -> Proxmox nic1/vmbr1: dedicated SPAN destination path for Security Onion capture.
- Cisco GigabitEthernet1/0/47 -> Windows 11 analyst station: VLAN 99 access port.
- Proxmox vmbr0 -> antiX VM with VLAN tag 10; Proxmox vmbr0 -> Security Onion management NIC with VLAN tag 99.
- Proxmox vmbr1 -> Security Onion capture NIC with no VLAN tag so mirrored frames arrive as raw capture traffic.

Reference Flow

Kali -> Netgear -> SRX ATTACKER zone -> SRX stateful policy -> SRX VICTIMS zone -> Cisco trunk -> VLAN 10 -> antiX target. In parallel, Cisco SPAN mirrors VLAN 10 in both directions to port 27 -> Proxmox nic1 -> vmbr1 -> Security Onion capture NIC -> analysis/SIEM pipeline.

5. Juniper SRX300 - Router-on-a-Stick and Stateful Boundary

The supplied Junos configuration uses vlan-tagging on ge-0/0/0 and creates logical units .10 and .99 for the VICTIMS and MGMT domains. The external attacker segment is placed directly on ge-0/0/5.0. Zone membership then becomes the basis for stateful policy enforcement.

5.1 Interface and gateway configuration

```
set interfaces ge-0/0/0 vlan-tagging
set interfaces ge-0/0/0.10 vlan-id 10
set interfaces ge-0/0/0.10 family inet address 172.16.10.1/24
set interfaces ge-0/0/0.99 vlan-id 99
set interfaces ge-0/0/0.99 family inet address 172.16.99.1/24
set interfaces ge-0/0/5 unit 0 family inet address 192.168.66.1/24
```

5.2 Zone assignment and host-inbound services

```
set security zones security-zone ATTACKER interfaces ge-0/0/5.0
set security zones security-zone ATTACKER host-inbound-traffic system-services ping

set security zones security-zone MGMT interfaces ge-0/0/0.99
set security zones security-zone MGMT host-inbound-traffic system-services all
set security zones security-zone MGMT host-inbound-traffic protocols all

set security zones security-zone VICTIMS interfaces ge-0/0/0.10
set security zones security-zone VICTIMS host-inbound-traffic system-services ping
```

5.3 Permit attacker-to-victim exploit traffic

```
set security policies from-zone ATTACKER to-zone VICTIMS policy ATTACK-TRAFFIC match source-address any
set security policies from-zone ATTACKER to-zone VICTIMS policy ATTACK-TRAFFIC match destination-address any
set security policies from-zone ATTACKER to-zone VICTIMS policy ATTACK-TRAFFIC match application any
```

```
set security policies from-zone ATTACKER to-zone VICTIMS policy ATTACK-TRAFFIC then permit
```

5.4 Management-to-attacker exception and explicit reverse deny

```
set security policies from-zone MGMT to-zone ATTACKER policy WIN-TO-KALI match source-address any
set security policies from-zone MGMT to-zone ATTACKER policy WIN-TO-KALI match destination-address any
set security policies from-zone MGMT to-zone ATTACKER policy WIN-TO-KALI match application any
set security policies from-zone MGMT to-zone ATTACKER policy WIN-TO-KALI then permit
```

```
set security policies from-zone ATTACKER to-zone MGMT policy BLOCK-KALI match source-address any
set security policies from-zone ATTACKER to-zone MGMT policy BLOCK-KALI match destination-address any
set security policies from-zone ATTACKER to-zone MGMT policy BLOCK-KALI match application any
set security policies from-zone ATTACKER to-zone MGMT policy BLOCK-KALI then deny
set security policies from-zone ATTACKER to-zone MGMT policy BLOCK-KALI then log session-init
commit
exit
```

Engineering rationale: the intended asymmetry is deliberate. Windows may initiate sessions toward Kali for laboratory control and testing; Kali is explicitly denied from initiating sessions into MGMT. This creates a measurable stateful boundary and a clean source of logged policy-deny telemetry.

6. Cisco Catalyst 2960-X - VLAN Trunking, Port Security, and SPAN

The switch is the physical Layer 2 enforcement and observation point. The supplied configuration creates VLANs 10, 99, and 999, disables unused ports into the blackhole VLAN, secures the analyst access port, trunks the Juniper and Proxmox links, and mirrors VLAN 10 in both directions to the Security Onion capture destination.

6.1 Baseline hardening and VLAN creation

```
hostname LAB-SW-01
ip domain-name lab.home
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
  transport input ssh
  login local

no ip http server
no ip http secure-server
no cdp run
vtp mode transparent

vlan 10
  name VICTIM_NETWORK
vlan 99
  name MANAGEMENT_NETWORK
vlan 999
  name BLACKHOLE
```

6.2 Management SVI, default gateway, and unused-port shutdown

```
interface Vlan99
  ip address 172.16.99.2 255.255.255.0
  no shutdown
  ip default-gateway 172.16.99.1

interface range GigabitEthernet1/0/2 - 26 , GigabitEthernet1/0/29 - 46 , GigabitEthernet1/0/48
```



```
description UNUSED_BLACKHOLE
switchport mode access
switchport access vlan 999
shutdown
```

6.3 Trunk ports

```
interface GigabitEthernet1/0/1
description UPLINK_TO_JUNIPER
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 999
no shutdown

interface GigabitEthernet1/0/27
description PROXMOX_MAIN_UPLINK
switchport mode trunk
switchport nonegotiate
spanning-tree portfast trunk
no shutdown
```

6.4 Analyst station access-port controls

```
interface GigabitEthernet1/0/47
description ANALYST_STATION_MGMT
switchport mode access
switchport access vlan 99
switchport nonegotiate
spanning-tree portfast
spanning-tree bpduguard enable
switchport port-security
switchport port-security maximum 1
switchport port-security violation restrict
switchport port-security mac-address sticky
no shutdown
```

6.5 SPAN destination and mirrored victim traffic

```
interface GigabitEthernet1/0/28
description SECONION_CAPTURE_NIC
switchport nonegotiate
no switchport port-security
no shutdown

no monitor session 1
monitor session 1 source vlan 10 both
monitor session 1 destination interface GigabitEthernet1/0/27

do write memory
```

Validation note: SPAN destination behavior is platform- and interface-mode dependent. The supplied configuration should be validated on the specific 2960-X hardware after deployment; the design intent is unambiguous - port 28 is dedicated to capture, not ordinary user traffic.

7. Proxmox VE - VLAN-Aware Fabric and Sensor Isolation

The Proxmox design deliberately uses two physical NICs. nic0 carries management and live VLAN-tagged traffic through vmbr0. nic1 carries only mirrored traffic through vmbr1. This separation prevents capture traffic from being mixed with the hypervisor management plane.

7.1 /etc/network/interfaces

```

auto lo
iface lo inet loopback

iface nic0 inet manual
iface nic1 inet manual

# CABLE 1: MANAGEMENT & LIVE TRAFFIC (Cisco Port 27)
auto vmbr0
iface vmbr0 inet static
    address 172.16.99.20/24
    gateway 172.16.99.1
    bridge-ports nic0
    bridge-stp off
    bridge-fd 0
    bridge-vlan-aware yes
    bridge-vids 2-4094

# CABLE 2: SPAN / CAPTURE TRAFFIC (Cisco Port 28)
auto vmbr1
iface vmbr1 inet manual
    bridge-ports nic1
    bridge-stp off
    bridge-fd 0
    bridge-ageing 0

source /etc/network/interfaces.d/*

```

7.2 Storage and NIC profile

Component	Supplied value	Engineering purpose
Management NIC (nic0)	fc:9d:05:05:87:6c	VLAN 99 management and VM traffic
Capture NIC (nic1)	6c:6e:07:50:e9:18	Dedicated SPAN ingress
External storage	1 TB SSD, ext4, SecOnion-Storage	High-I/O Security Onion storage
Web GUI	https://172.16.99.20:8006	Proxmox administration

7.3 Virtual machine bindings

- antiX target node(s): storage on local-lvm; net0 -> vmbr0; VLAN tag 10; Proxmox firewall disabled so raw lab exploit traffic reaches the guest.
- Security Onion 3.1.0: storage on SecOnion-Storage; management net0 -> vmbr0 with VLAN 99; Proxmox firewall checked on management NIC; capture net1 -> vmbr1 with no VLAN tag; Proxmox firewall unchecked on the capture NIC.

8. Security Onion 3.1.0 and Telemetry Path

The sensor has two logically separate roles: management and capture. The management interface participates in VLAN 99 for administrative access. The capture interface is attached to vmbr1 and receives the Cisco SPAN feed without a VLAN tag so the sensor can process mirrored frames as raw traffic.

The supplied topology describes this as an out-of-band telemetry path: the victim traffic is mirrored at the physical switch, passes into Proxmox through the dedicated capture NIC, and is presented to Security Onion independently of the production-like routing path.

Detection / Threat-Hunting POC

The supplied Splunk searches focus on DNS records with query strings longer than 30 characters. This is a lightweight heuristic for finding potential encoded or data-bearing DNS activity, then ranking source IPs by volume and preserving representative long queries for triage.

8.1 POC query - extremes / top talker view

```
index=data_exfil sourcetype="dns_logs"
| eval long_query=if(len(query) > 30, query, null())
| stats count values(long_query) as long_queries by src_ip
| eval long_queries=mvindex(long_queries, 0, 4)
| eventstats max(count) as max_count, min(count) as min_count
| where count=max_count OR count=min(count)
| fields - max_count, min_count
| sort -count
```

8.2 POC query - top / bottom IP ranking

```
index=data_exfil sourcetype="dns_logs"
| eval exfil_query=if(len(query) > 30, query, null())
| stats count values(exfil_query) as dns_exfil by src_ip
| eval dns_exfil=mvindex(dns_exfil, 0, 4)
| sort - count
| streamstats count as top_ten_ips
| sort + count
| streamstats count as bottom_ten_ips
| where top_ten_ips <= 10 OR bottom_ten_ips <= 10
| eval Category=case(top_ten_ips <= 10, "Top 10 IPs", bottom_ten_ips <= 10, "Bottom 10 IPs")
| fields - top_ten_ips, bottom_ten_ips
| sort - count
| table Category, src_ip, count, dns_exfil
```

Analytic interpretation: the lab can be used to produce a controlled DNS-exfiltration-like data set, then validate whether the sensor and SIEM preserve the source IP, query string, and event frequency required for analyst triage. The queries are heuristics, not proof of exfiltration; thresholding and baseline should be tuned to the environment.

9. Kali Linux - Attacker Network Preparation

Kali is statically anchored at 192.168.66.50/24 with the Juniper attacker gateway 192.168.66.1. The source procedure first flushes the interface to remove conflicting addresses, then defines a NetworkManager static profile.

9.1 Static attacker profile

```
sudo ip addr flush dev eth0

sudo nmcli connection modify "Wired connection 1" ipv4.method manual ipv4.addresses 192.168.66.50/24 ipv4.gateway 192.168.66.1

sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

9.2 Verification

```
ip a
ip route
ping 172.16.10.1
```

The expected logical result is a clean attacker interface, a default gateway of 192.168.66.1, and successful reachability to the victim gateway only because the SRX permits the relevant zone-to-zone flow.

10. Windows 11 Analyst Station - Dual-Homed Command Center

The Windows station is designed to use Wi-Fi for general Internet access while Ethernet remains dedicated to the isolated management plane. The Ethernet interface receives a static address but no default gateway. Explicit persistent routes are used to reach the broader lab and attacker subnet through the SRX management gateway.

10.1 Addressing and persistent routing

```
# Run in elevated PowerShell
New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 172.16.99.10 -PrefixLength 24

# Persistent lab routes
route -p add 172.16.0.0 mask 255.255.0.0 172.16.99.1
route -p add 192.168.66.0 mask 255.255.255.0 172.16.99.1
```

The explicit route design prevents the lab Ethernet interface from becoming a general Internet path while still allowing administrative access to VLAN 99, the victim subnet, and the attacker network.

10.2 Legacy SSH compatibility - scoped exception

The source notes explain that current Windows OpenSSH may reject the older key-exchange and host-key algorithms still offered by the lab equipment. The workaround is a host-specific ~/.ssh/config exception rather than a global SSH downgrade.

10.3 Host-specific SSH compatibility file

```
$sshDir = "$env:USERPROFILE\.ssh"
if (-not (Test-Path $sshDir)) { New-Item -ItemType Directory -Path $sshDir }
$configContent = @"
# Juniper SRX300
Host 172.16.99.1
    KexAlgorithms +diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1
    HostKeyAlgorithms +ssh-rsa

# Cisco Catalyst 2960-X
Host 172.16.99.2
    KexAlgorithms +diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1
```

```
HostKeyAlgorithms +ssh-rsa
"@
Set-Content -Path "$sshDir\config" -Value $configContent -Encoding Ascii
```

Security note: SHA-1-based SSH algorithms are intentionally weaker and should not be enabled globally. In this lab they exist only as a compatibility exception for legacy equipment, and the exception should be removed when the equipment supports modern algorithms.

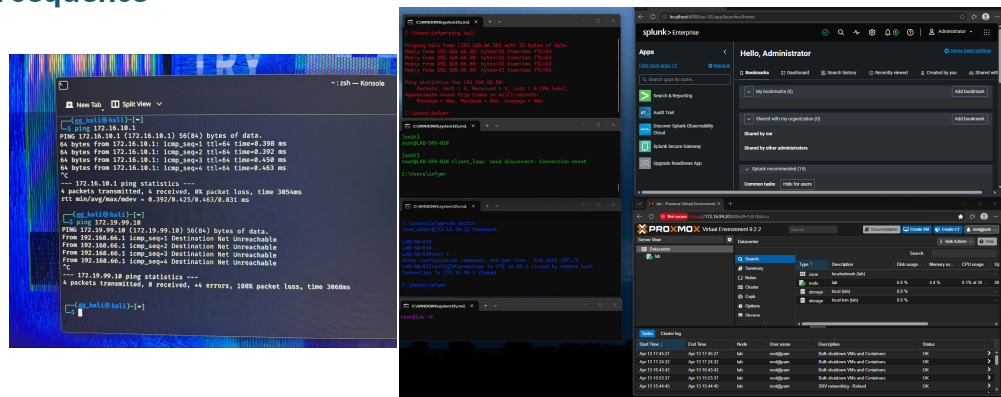
10.4 Local name resolution

10.4 Static hosts-file aliases

```
# C:\Windows\System32\drivers\etc\hosts
172.16.99.20 lab.home lab proxmox
172.16.99.1 firewall.home firewall srx juniper
172.16.99.2 switch.home switch cisco
172.16.99.30 seconion.home seconion so
192.168.66.50 kali.home kali attacker
172.16.10.15 antix.home antix victim target
```

10.5 Analyst validation sequence

- ping firewall
- ping lab
- ping 172.16.10.1
- ping antix
- ssh soc_admin@switch
- ssh root@firewall
- ping kali



The source checklist notes that the Kali ping may time out when the Kali endpoint is powered off; that is a useful test of physical availability rather than a configuration failure.

11. End-to-End Build Runbook

11.1 Inventory and cabling

Connect the physical interfaces exactly as shown in the topology: attacker switch to SRX ge-0/0/5; SRX ge-0/0/0 to Cisco Gi1/0/1; Cisco Gi1/0/27 to Proxmox nic0; Cisco Gi1/0/28 to Proxmox nic1; Cisco Gi1/0/47 to Windows 11.

11.2 Build the SRX routing plane

Create VLAN-tagged subinterfaces .10 and .99, assign 172.16.10.1/24 and 172.16.99.1/24, then bind the interfaces to VICTIMS and MGMT zones. Add the attacker interface 192.168.66.1/24 and place it in ATTACKER.

11.3 Enforce zone policy

Permit ATTACKER -> VICTIMS via ATTACK-TRAFFIC. Permit MGMT -> ATTACKER for operator-initiated testing. Explicitly deny ATTACKER -> MGMT and log session initiation.

11.4 Configure Cisco

Create VLANs 10/99/999, harden management, disable unused ports, trunk Gi1/0/1 and Gi1/0/27, configure Gi1/0/47 as a locked-down management access port, then create the VLAN 10 SPAN session to Gi1/0/28.

11.5 Configure Proxmox

Assign nic0 and nic1, create vmbr0 as VLAN-aware with 172.16.99.20/24 and gateway 172.16.99.1, and create vmbr1 as a capture-only bridge with no IP configuration.

11.6 Bind VMs

Place antiX on vmbr0 with VLAN 10. Place Security Onion management on vmbr0 with VLAN 99. Place Security Onion capture on vmbr1 with no VLAN tag.

11.7 Configure Kali

Flush the attacker NIC, set 192.168.66.50/24 and gateway 192.168.66.1, bounce NetworkManager, and verify routing.

11.8 Configure Windows

Assign 172.16.99.10/24 to the Ethernet interface without a gateway, add persistent routes, create the scoped SSH compatibility file, and add local host aliases.

11.9 Validate the telemetry chain

Generate controlled attacker traffic toward the antiX target. Confirm the SRX session is permitted, the Cisco SPAN session counters increment, Security Onion sees the frames, and Splunk receives the expected records.

11.10 Run threat-hunting queries

Use the supplied DNS query heuristics to identify long DNS queries, rank source IPs, preserve representative query values, and compare the observed results with the known lab-generated activity.

12. Proof-of-Concept Acceptance Criteria

Test	Expected result	Evidence to capture
Windows -> firewall	Successful management reachability	ping output / SSH session
Windows -> antiX	Route through SRX to VLAN 10 succeeds	ping 172.16.10.15
Windows -> Kali	Operator-initiated access succeeds	SSH / ping / session log
Kali -> antiX	Attack path is permitted	SRX session log + packet capture
Kali -> Windows	Initiation is explicitly denied and logged	SRX BLOCK-KALI log
Cisco SPAN	VLAN 10 traffic is mirrored both directions	SPAN counters + packet capture
Security Onion capture	Raw frames arrive on the capture NIC	tcpdump / sensor telemetry
Splunk DNS analytics	Long DNS query events are ranked and	Query output with source IP and query

13. Troubleshooting Matrix

Symptom	Likely control plane	Check
Windows cannot reach 172.16.10.15	Static routing / SRX MGMT-VICTIMS path	route print; ping 172.16.10.1; confirm VLAN 99 gateway
Kali cannot reach 172.16.10.1	Attacker interface or SRX policy	ip a; ip route; check ATTACKER->VICTIMS policy
AntiX receives no exploit traffic	VLAN tagging / Proxmox firewall	Confirm VM tag 10 and Proxmox firewall disabled for target
Security Onion capture is empty	SPAN / vmbri / capture NIC	Confirm monitor session 1; port 28; vmbri has no IP and raw frames
SSH to Juniper/Cisco fails	Legacy algorithm mismatch	Confirm host-specific ~/.ssh/config exception
Proxmox unreachable	VLAN 99 / vmbri / gateway	Check 172.16.99.20/24, vmbri VLAN-aware, Cisco Gi1/0/27 trunk
DNS hunting query returns no data	Ingestion/index/sourcetype mismatch	Confirm index=data_exfil and sourcetype=dns_logs

14. Security Engineering Observations

- Explicitly deny attacker-initiated access into MGMT and log session initiation so the POC demonstrates prevention plus telemetry.
- Do not carry management credentials into a public repository or publication. The source notes contain real-looking secrets; rotate them before any external release.
- Keep legacy SSH algorithm exceptions host-specific and temporary. Prefer firmware/OS upgrades that support modern cryptographic suites.
- Keep the capture NIC separate from the management NIC. A capture bridge should not be assigned an address unless there is a deliberate reason to do so.
- Use VLAN 999 as a blackhole/parking VLAN for unused switch access ports, and keep unused ports administratively down as shown in the supplied configuration.
- Treat the antiX VM as an intentionally vulnerable target. Disable unnecessary outbound access and reset/restore the snapshot between attack experiments.
- Use the SPAN source definition narrowly. Mirroring an entire VLAN is useful for POC visibility, but production-scale deployments require bandwidth and packet-loss considerations.

16. Source Material and Scope

- Windows Configurations(1).md - Windows dual-homed networking, persistent routes, SSH compatibility, hosts-file aliases, validation commands.
- Proxmox settings.md - vmbr0/vmbr1 design, NIC bindings, storage, VM network settings.
- Logins&Password.md - lab access details; operational credentials intentionally omitted from this public edition.
- Kali Config.md - attacker NIC reset, static profile, route verification.
- Juniper Configuration.md - SRX router-on-a-stick, zones, policies, and logging.
- IPs.md - trust-zone and subnet mapping.
- Cisco Config.txt - VLANs, trunk ports, management access, port security, SPAN, and SSH compatibility.
- Topology.jpg - authoritative physical/virtual topology diagram supplied with the lab.

17. Appendix A - Public Credential Handling

The source notes contain usernames and passwords for Splunk, Proxmox, Juniper, Kali, and Cisco. They are not reproduced here. For an actual private runbook, maintain the credential matrix in a password manager or encrypted secret store and reference it from the lab document by system name rather than embedding secrets in the build guide.

Public release checklist: rotate any credentials that have been committed to a file, repository, screenshot, or previous publication; remove secrets from version control history where applicable; verify screenshots for password leakage; and keep the public PDF focused on architecture, engineering decisions, evidence, and repeatability.